Type MS · State or explain safety precautions for a practical

Answer 1 of 11 · Question 2 part a.ii · 2 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

number			
2 (a) (i)	better transfer of heat to water (1)		1
(a) (ii)	any two from • safety goggles/glasses (1) • water may spit (1) • use test tube holder (1) • to avoid burning hands (1)	reason linked to precaution	2
(b) (i)	• 28 x 12 x 4.2 (1) • 1411/1411.2J (1) • 1400J (1)	error carried forward full marks for correct answer	3
(ii)	• some energy lost to air/absorbed by tube/needle (1) • some converted to light (1) • food not fully burnt;		3
(c)	• use same mass of food (1) • same volume of water (1) • use same thin tube (1) • burn food same distance from tube (1)		4

Total 13 marks

Type MS · State or explain safety precautions for a practical

Answer 1 of 11 · Question 2 part a.ii · 2 marks

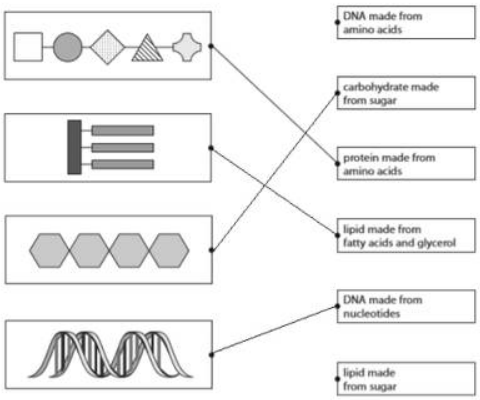
2020 NOV · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State or explain safety precautions for a practical

Answer 2 of 11 · Question 3 part d · 2 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part d in this mark scheme extract

3	(a)	( ) 	Do not allow more than one line from each structure	3
	(b)	**D** stomach;		1
	(c) (i)	3/three		1
	(ii)	not washing equipment/transfer of food containing protein into distilled water;	Allow any valid alternative	1
	(d)	Hazard: Broken glass - cuts Reducing risk: Clear up breakages using a dustpan and brush/keep equipment away from edge of tables;	Ignore wear gloves	1
		Hazard: Biuret reagent - irritant Reducing risk: Wear goggles/wash hands after use/don't use if skin is sensitive;	Ignore wear gloves	1

Total question 3 = 8 marks

Type MS · State or explain safety precautions for a practical

Answer 2 of 11 · Question 3 part d · 2 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part d in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State or explain safety precautions for a practical

Answer 3 of 11 · Question 6 part b.ii · 1 mark

2020 JAN · Paper 01 · find part b.ii in this mark scheme extract

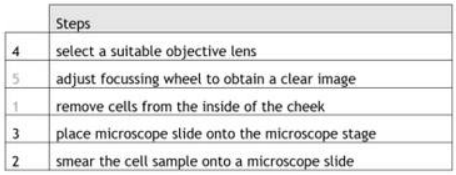
Number				
6	(a)	bronchus/bronchi (1)		1
	(b) (i)	• inhale and exhale/breathe in an out (1) • through mouthpiece (1) • floating air chamber rises/falls (1) • pen records a trace on (the rotating) graph paper/drum (1)		4
	(ii)	Do not share the mouthpiece with another person/disinfect mouthpiece before use by another person/after use (1)		1
	(iii)	Any two from: • airways /bronchi/bronchus constricted/obstructed/reduced diameter (1) • less/reduced airflow (1) • per unit time (1)		2
	(c) (i)	cannot be transmitted from one person to another/not contagious/not caused by a pathogen/bacteria/virus/microorganism (1)		1
	(ii)	9.5 - 7 = 2.5(1) 2.5 ÷ 7 x 100(1) 36(%) (1)	Full marks for correct final answer Allow 35.7 for final answer	3
	(iii)	people taking less exercise/ consuming too much fat/sugar/ taking in more energy than is being used/ increase in the number of people smoking/drinking more alcohol/obesity (1)	Allow other valid lifestyle factor	1

Total for Question 6 = 13 marks

Type MS · State or explain safety precautions for a practical

Answer 4 of 11 · Question 3 part a.iv · 2 marks

2020 NOV · Paper 01 (regional R) · find part a.iv in this mark scheme extract

Number				
3	(a) (i)		One mark for 4 before 5. One mark for 3 before 2	2
	(ii)	more detail/organelles can be seen/organelles/named organelle seen more clearly (1)		1
	(iii)	reference no stain applied to specimen(1)		1
	(iv)	• reference to sterilising (named) equipment (1) • wash hands to avoid cross contamination(1) • incubate/store in sealed petri dish / container	allow valid sterilisation technique	2
	(b) (i)	10 × 40 = 400	Full marks for correct answer without working	1
	(ii)	reference to 1000 (1) 0.018 × 1000 = 18 (µm)	Full marks for correct final answer	2

Total Question 3 = 9 marks

Type MS · State or explain safety precautions for a practical

Answer 5 of 11 · Question 7 part a.i · 3 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

number				
7	(a) (i)	• safety glasses/goggles; • care using sharp needle; • care to avoid burns; •		3
	(ii)	• mass of food; • volume of water; • distance of burning food from tube;	reject amount	3
	(iii)	any five from • not all food burns to release energy; • not all heat transferred to water; • some lost to atmosphere; • some heat lost to needle/glass tube; • uneven distribution of temperature in water; • because lack of stirring;		5
	(b)	any four from • use of oxygen; • allows complete combustion; • enclosed/lid so no heat escapes; • heat transfer coil so all heat is used to increase temp of water/transferred to water; • stirrer to distribute heat;		4

Total 15 marks

Type MS · State or explain safety precautions for a practical

Answer 5 of 11 · Question 7 part a.i · 3 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State or explain safety precautions for a practical

Answer 6 of 11 · Question 3 part a.iii · 2 marks

2022 JUN · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

3 (a) (i)	Any five from the following: • place water in the test tube; • take the (starting) temperature of the water; • attach the test tube to a clamp stand; • measure the mass of a piece of food/weigh food; • place food on the mounted needle (under the test tube); • burn the food; (using a Bunsen burner) • take the (final) temperature of the water;		5
(ii)	Any two from the following: • volume/mass of water; • distance of the food from the test tube; • mass of food;	ignore amount ignore amount/size	2
(iii)	Any two from the following: • hold the burning food at arm's length; • reference to sharpness of needle; • goggles; • reference to food allergies/don't taste food;	any ref to being burnt/hot water	2
(b) (i)	Any two from the following: • males need more energy than females; • the amount of energy required decreases with age; • the amount of energy needed by males decreases more (overall) than females;		2
(ii)	**D** - Males use more muscle mass than females;		1
(iii)	• (2629 – 2103) = 526; • 526 ÷ 2103; • (x 100) = 25 (%);	if divide by 2629 and give 20% then two marks Total 15 marks	3

Type MS · State or explain safety precautions for a practical

Answer 6 of 11 · Question 3 part a.iii · 2 marks

2022 JUN · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State or explain safety precautions for a practical

Answer 7 of 11 · Question 1 part d · 1 mark

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part d in this mark scheme extract

1 (a)	• apparatus in correct order with T piece connected to both flasks; • long tubes in indicator; • short tubes above indicator;		3
(b)	limewater;	Allow sodium hydrogen carbonate	1
(c)	any four from: • breathe through mouth piece; • in and out; • 10 times/max of one minute; • compare contents of flasks; • exhaled flask should go cloudy if more carbon dioxide exhaled;		4
(d)	sterilise mouth piece/breath gently;		1
(e)	same volume of indicator in each flask;		1

Total 10 marks

Type MS · State or explain safety precautions for a practical

Answer 7 of 11 · Question 1 part d · 1 mark

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part d in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State or explain safety precautions for a practical

Answer 8 of 11 · Question 1 part c.i · 1 mark

2024 JUN · Paper 01 · find part c.i in this mark scheme extract

1 (a)	(i)	sensory (neurone/cell);		(1)
	(ii)	axon;		(1)
(b)	(i)	Part R sweat gland; Part S blood vessels/capillaries/arterioles;		(2)
	(ii)	Any two from: • produces sweat/perspiration; • sweat evaporates (from skins surface); • using heat energy (from the blood);		(2)
	(iii)	• provides an insulating layer insulation; • traps heat (energy)/to prevent heat loss;	reject keeps body warm	(2)
(c)	(i)	• water is too hot/burn hands;		(1)
	(ii)	• use a thermometer; • to test the temperature of the water/to make sure the water is not too hot;		(2)
	(iii)	• (temperature) receptor;		(1)

Total for Question 1 = 12 Marks

Type MS · State or explain safety precautions for a practical

Answer 9 of 11 · Question 1 part c.ii · 2 marks

2024 JUN · Paper 01 · find part c.ii in this mark scheme extract

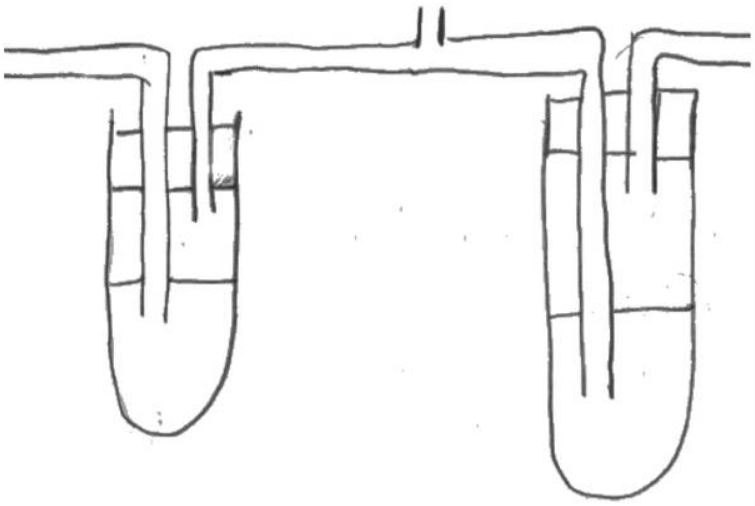
1 (a)	(i)	sensory (neurone/cell);		(1)
	(ii)	axon;		(1)
(b)	(i)	Part R sweat gland; Part S blood vessels/capillaries/arterioles;		(2)
	(ii)	Any two from: • produces sweat/perspiration; • sweat evaporates (from skins surface); • using heat energy (from the blood);		(2)
	(iii)	• provides an insulating layer insulation; • traps heat (energy)/to prevent heat loss;	reject keeps body warm	(2)
(c)	(i)	• water is too hot/burn hands;		(1)
	(ii)	• use a thermometer; • to test the temperature of the water/to make sure the water is not too hot;		(2)
	(iii)	• (temperature) receptor;		(1)

Total for Question 1 = 12 Marks

Type MS · State or explain safety precautions for a practical

Answer 10 of 11 · Question 4 part a.ii · 1 mark

2024 JUN · Paper 01 · find part a.ii in this mark scheme extract

4	(a)	(i)	• apparatus correctly set up; • long tube in indicator; • short tube above indicator; 		(3)
		(ii)	Inhale and exhale gently/sterilise/clean mouthpiece/don't inhale liquid/care in assembling apparatus to avoid breaking glass;		(1)
	(b)	(i)	exercise/rest for the same length of time/at the same pace for each repeat/same student;		(1)
		(ii)	any four from • breathing rate increases (with vigorous exercise); • more oxygen inhaled/supplied; • for aerobic respiration; • more energy released/more ATP; • breakdown of lactic acid; • produced by anaerobic respiration;	ignore ref CO_2	(4)

Total for Question 4 = 9 marks

Type MS · State or explain safety precautions for a practical

Answer 11 of 11 · Question 8 part b.ii · 1 mark

2025 JUN · Paper 02 (regional R) · find part b.ii in this mark scheme extract

8 (a)	(i)	cilia/skin/mucus	allow other suitable answers	1
	(ii)	kills/inhibits growth of pathogens/damage cell membranes;		1
	(iii)	• (phagocytes) engulf pathogens/microorganisms / bacteria; • digest/breakdown pathogens/microorganisms / bacteria;		2
	(iv)	antigen/immunogen;		1
	(v)	• antibodies have a particular shape /are complementary (to antigen); • only recognise/attach/bind to, one particular/specific pathogen / antigen/receptor;		2
	(vi)	• natural passive immunity the body receives antibodies naturally;	accept through breast milk/placenta	
		• artificial passive immunity are when antibodies are injected into the body;	accept e.g. to give immediate protection against a disease;	2
(b)	(i)	360 ÷ 30 = 12; 108 ÷ 12; 9;		3
	(ii)	reduce the risk of disease transmission/infection;		1
	(iii)	any two from • improper/overuse of antibiotics/over prescribed; • bacterial DNA mutates/bacteria adapt; • some survive and reproduce		2
			Total for Question 8 = 15 marks	